

CHAPTER 11

Thermodynamics

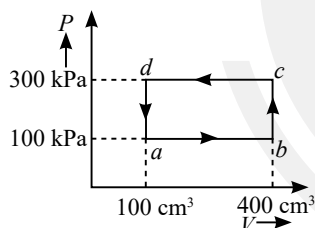
Thermodynamic Processes and First Law of Thermodynamics

1. The equilibrium state of a thermodynamic system is described by

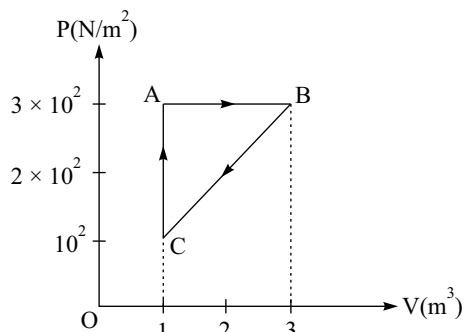
A. Pressure
B. Total heat
C. Temperature
D. Volume
E. Work done

Choose the most appropriate answer from the options given below. (2024 Re)

- a. A, B and E only
b. B, C and D only
c. A, B and C only
d. A, C and D only
2. A thermodynamic system is taken through the cycle $abcda$. The work done by the gas along the path bc is: (2024)

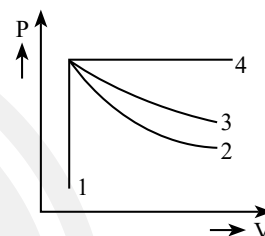


- a. -90J
b. -60J
c. Zero
d. 30J
3. For the given cycle, the work done isobaric process is: (2023-Manipur)

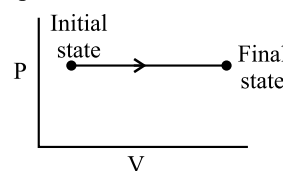


- a. 200J
b. Zero
c. 400J
d. 600J
4. An ideal gas follows a process described by the equation $PV^2 = C$ from the initial (P_1, V_1, T_1) to final (P_2, V_2, T_2) thermodynamic states, where C is a constant. Then: (2022 Re)
- a. If $P_1 > P_2$ then $V_1 > V_2$
b. If $P_1 > P_2$ then $T_1 < T_2$
c. If $V_2 > V_1$ then $T_2 > T_1$
d. If $V_2 > V_1$ then $T_2 < T_1$

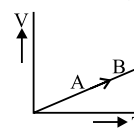
5. An ideal gas undergoes four different processes from the same initial state as shown in the figure below. Those processes are adiabatic, isothermal isobaric and isochoric. The curve which represents the adiabatic process among 1, 2, 3 and 4 is: (2022)



- a. 4
b. 1
c. 2
d. 3
6. Two cylinders A and B of equal capacity are connected to each other via a stop cock. A contains an ideal gas at standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. The stop cock is suddenly opened. The process is: (2020)
- a. Adiabatic
b. Isochoric
c. Isobaric
d. Isothermal
7. The P-V diagram for an ideal gas in a piston cylinder assembly undergoing a thermodynamic process is shown in the figure. The process is (2020-Covid)



- a. Isochoric
b. Isobaric
c. Isothermal
d. Adiabatic
8. In which of the following processes, heat is neither absorbed nor released by a system? (2019)
- a. Isothermal
b. Adiabatic
c. Isobaric
d. Isochoric
9. The volume (V) of a monoatomic gas varies with its temperature (T), as shown in the graph. The ratio of the work done by the gas, to the heat absorbed by it, when it undergoes a change from state A to State B, is (2018)

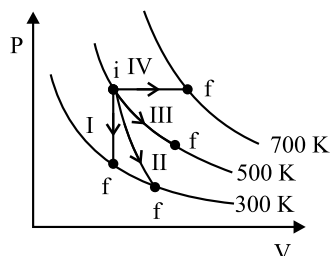


- a. $\frac{1}{3}$
b. $\frac{2}{3}$
c. $\frac{2}{5}$
d. $\frac{2}{7}$

10. A sample of 0.1 g of water at 100°C and normal pressure ($1.013 \times 10^5 \text{ Nm}^{-2}$) requires 54 cal of heat energy to convert to steam at 100°C . If the volume of the steam produced is 167.1 cc, the change in internal energy of the sample, is: (2018)

a. 42.2 J b. 208.7 J c. 104.3 J d. 84.5 J

11. Thermodynamic processes are indicated in the following diagram. (2017-Delhi)



Match the following:

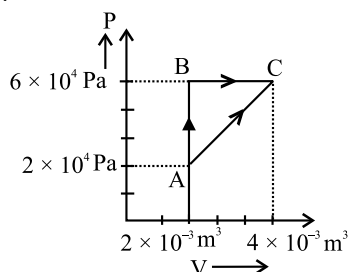
	Column-1		Column-2
P.	Process I	a.	Adiabatic
Q.	Process II	b.	Isobaric
R.	Process III	c.	Isochoric
S.	Process IV	d.	Isothermal

- a. $P \rightarrow c$, $Q \rightarrow a$, $R \rightarrow d$, $S \rightarrow b$
 b. $P \rightarrow c$, $Q \rightarrow d$, $R \rightarrow b$, $S \rightarrow a$
 c. $P \rightarrow d$, $Q \rightarrow b$, $R \rightarrow a$, $S \rightarrow c$
 d. $P \rightarrow a$, $Q \rightarrow c$, $R \rightarrow d$, $S \rightarrow b$

12. A gas is compressed isothermally to half its initial volume. The same gas is compressed separately through an adiabatic process until its volume is again reduced to half. Then: (2016 - I)

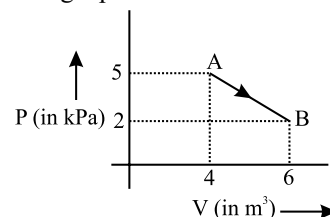
- a. Compressing the gas isothermally will require more work to be done
 b. Compressing the gas through adiabatic process will require more work to be done
 c. Compressing the gas isothermally or adiabatically will require the same amount of work
 d. Which of the case (whether compression through isothermal or through adiabatic process) requires more work will depend upon the atomicity of the gas

13. Figure below shows two paths that may be taken by a gas to go from a state A to a state C. In process AB, 400 J of heat is added to the system and in process BC, 100 J of heat is added to the system. The heat absorbed by the system in the process AC will be: (2015)



a. 500 J b. 460 J c. 300 J d. 380 J

14. One mole of an ideal diatomic gas undergoes a transition from A to B along a path AB as shown in the figure,



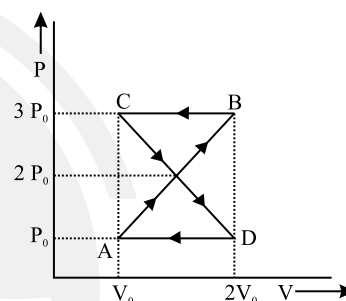
The change in internal energy of the gas during the transition is: (2015)

a. -20 kJ b. 20 J c. -12 kJ d. 20 kJ

15. An ideal gas is compressed to half its initial volume by means of several processes. Which of the process results in the maximum work done on the gas? (2015 Re)

a. Isothermal b. Adiabatic c. Isobaric d. Isochoric

16. A thermodynamic system undergoes cyclic process ABCDA as shown in figure. The work done by the system in the cycle is: (2014)



a. $P_0 V_0$ b. $2 P_0 V_0$ c. $\frac{P_0 V_0}{2}$ d. Zero

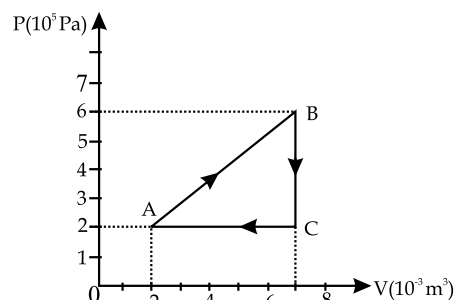
17. A monoatomic gas at a pressure P , having a volume V expands isothermally to a volume $2V$ and then adiabatically to a volume $16V$. The final pressure of the gas is (take $\gamma = 5/3$): (2014)

a. $64P$ b. $32P$ c. $P/64$ d. $16P$

18. During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its temperature. The ratio of $\frac{C_p}{C_v}$ for the gas is: (2013)

a. $3/2$ b. 2 c. $4/3$ d. $5/3$

19. A gas is taken through the cycle $A \rightarrow B \rightarrow C \rightarrow A$, as shown. What is the net work done by the gas? (2013)



a. -2000 J b. 2000 J c. 1000 J d. Zero

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (d) | 2. (c) | 3. (d) | 4. (d) | 5. (c) | 6. (a) | 7. (b) | 8. (b) | 9. (c) | 10. (b) |
| 11. (a) | 12. (b) | 13. (b) | 14. (a) | 15. (b) | 16. (d) | 17. (c) | 18. (a) | 19. (c) | |

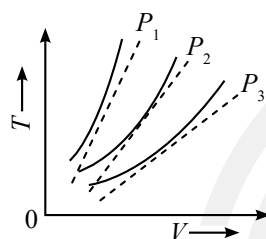


CHAPTER 12

Kinetic Theory

Ideal Gas Equations and Vander Waals Relation

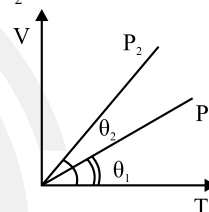
1. The following graph represents the T - V curves of an ideal gas (where T is the temperature and V the volume) at three pressures P_1 , P_2 and P_3 compared with those of Charles's law represented as dotted lines. (2024)



- a. $P_2 > P_1 > P_3$ b. $P_1 > P_2 > P_3$
c. $P_3 > P_2 > P_1$ d. $P_1 > P_3 > P_2$
2. The volume occupied by the molecules contained in 4.5 kg water at STP, if the intermolecular forces **vanish away** is: (2022)
- a. 5.6 m^3 b. $5.6 \times 10^6 \text{ m}^3$
c. $5.6 \times 10^3 \text{ m}^3$ d. $5.6 \times 10^{-3} \text{ m}^3$
3. A cylinder contains hydrogen gas at pressure of 249 kPa and temperature 27°C . Its density is : ($R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$) (2020)
- a. 0.2 kg/m^3 b. 0.1 kg/m^3 c. 0.02 kg/m^3 d. 0.5 kg/m^3
4. An ideal gas equation can be written as $P = \frac{\rho RT}{M_0}$ where ρ and M_0 are respectively, (2020-Covid)
- a. Number density, molar mass
b. Mass density, molar mass
c. Number density, mass of the gas
d. Mass density, mass of the gas
5. Increase in temperature of a gas filled in a container would lead to: (2019)
- a. Increase in its mass
b. Increase in its kinetic energy
c. Decrease in its pressure
d. Decrease in intermolecular distance
6. A given sample of an ideal gas occupies a volume V at a pressure P and absolute temperature T . The mass of each molecule of the gas is m . Which of the following gives the density of the gas? (2016 - II)
- a. $P/(kTV)$ b. mkT c. $P/(kT)$ d. $Pm/(kT)$

7. Two vessels separately contain two ideal gases A and B at the same temperature, the pressure of A being twice that of B. Under such conditions, the density of A is found to be 1.5 times the density of B. The ratio of molecular weight of A and B is: (2015 Re)
- a. $1/2$ b. $2/3$ c. $3/4$ d. 2

8. In the given (V - T) diagram, what is the relation between pressures P_1 and P_2 ? (2013)



- a. Cannot be predicted b. $P_2 = P_1$
c. $P_2 > P_1$ d. $P_2 < P_1$

Pressure Exerted by an Ideal Gas and its Kinetic Energy

9. A container of volume 200 cm^3 contains 0.2 mole of hydrogen gas and 0.3 mole of argon gas. The pressure of the system at temperature 200 K ($R = 8.3 \text{ JK}^{-1} \text{ mol}^{-1}$) will be: (2023-Manipur)
- a. $6.15 \times 10^5 \text{ Pa}$ b. $6.15 \times 10^4 \text{ Pa}$
c. $4.15 \times 10^5 \text{ Pa}$ d. $4.15 \times 10^6 \text{ Pa}$

Speed of Gas Molecules

10. The temperature of a gas is -50°C . To what temperature the gas should be heated so that the rms speed is increased by 3 times? (2023)
- a. 223 K b. 669°C c. 3295°C d. 3097 K
11. Three vessels of equal capacity have gases at the same temperature and pressure. The first vessel contains helium (monoatomic), the second contains fluorine (diatomic) and the third contains sulfur hexafluoride (polyatomic). The correct statement, among the following is: (2022 Re)
- a. The root mean square speed of sulfur hexafluoride is the largest
b. All vessels contain unequal number of respective molecules
c. The root mean square speed of molecules is same in all three cases
d. The root mean square speed of helium is the largest

12. Match Column-I and Column-II and choose the correct match from the given choices. (2021)

Column-I	Column-II
(A) Root mean square speed of gas molecules	(P) $\frac{1}{3}nm\bar{v}^2$
(B) Pressure exerted by ideal gas	(Q) $\sqrt{\frac{3RT}{M}}$
(C) Average kinetic energy of a molecule	(R) $\frac{5}{2}RT$
(D) Total internal energy of 1 mole of a diatomic gas	(S) $\frac{3}{2}k_B T$

- a. (A) - (Q), (B) - (R), (C) - (S), (D) - (P)
 b. (A) - (Q), (B) - (P), (C) - (S), (D) - (R)
 c. (A) - (R), (B) - (Q), (C) - (P), (D) - (S)
 d. (A) - (R), (B) - (P), (C) - (S), (D) - (Q)

13. At what temperature will the rms speed of oxygen molecules become just sufficient for escaping from the Earth's atmosphere? (Given: Mass of oxygen molecule (m) = 2.76×10^{-26} kg, Boltzmann's constant $k_B = 1.38 \times 10^{-23}$ JK⁻¹) (2018)

- a. 5.016×10^4 K b. 8.360×10^4 K
 c. 2.508×10^4 K d. 1.254×10^4 K

14. The molecules of a given mass of a gas have r.m.s velocity of 200 ms^{-1} at 27°C and $1.0 \times 10^5 \text{ Nm}^{-2}$ pressure. When the temperature and pressure of the gas are respectively, 127°C and $0.05 \times 10^5 \text{ Nm}^{-2}$, the r.m.s. velocity of its molecules in ms^{-1} is: (2016 - I)

- a. $100\sqrt{2}$ b. $\frac{400}{\sqrt{3}}$ c. $\frac{100\sqrt{2}}{3}$ d. $\frac{100}{3}$

Variation of Pressure with Depth and Pascal's Law

15. The average thermal energy for a mono-atomic gas is : (k_B is Boltzmann constant and T is absolute temperature) (2020)

- a. $\frac{3}{2} k_B T$ b. $\frac{5}{2} k_B T$ c. $\frac{7}{2} k_B T$ d. $\frac{1}{2} k_B T$

16. A gas mixture consists of 2 moles of O_2 and 4 moles of Ar at temperature T . Neglecting all vibrational modes, the total internal energy of the system is: (2017-Delhi)

- a. $15 RT$ b. $9 RT$ c. $11 RT$ d. $4 RT$

17. The molar specific heats of an ideal gas at constant pressure and volume are denoted by C_p and C_v respectively. If $\gamma = \frac{C_p}{C_v}$ and R is the universal gas constant, then C_v is equal to: (2013)

- a. γR b. $\frac{1+\gamma}{1-\gamma}$ c. $\frac{R}{(\gamma-1)}$ d. $\frac{(\gamma-1)}{R}$

Specific and Molar Heat Capacity of Gases

18. According to the law of equipartition of energy, the number of vibrational modes of a polyatomic gas of constant $\gamma = \frac{C_p}{C_v}$

is (C_p where C_v are the specific heat capacities of the gas at constant pressure and constant volume, respectively):

(2024 Re)

- a. $\frac{4+3\gamma}{\gamma-1}$ b. $\frac{3+4\gamma}{\gamma-1}$ c. $\frac{4-3\gamma}{\gamma-1}$ d. $\frac{3-4\gamma}{\gamma-1}$

19. One mole of an ideal monoatomic gas undergoes a process described by the equation $PV^3 = \text{constant}$. The heat capacity of the gas during this process is: (2016 - II)

- a. $2 R$ b. R c. $\frac{3}{2} R$ d. $\frac{5}{3} R$

20. The ratio of the specific heats $\frac{C_p}{C_v} = \gamma$ in terms of degrees of freedom (n) is given by: (2015)

- a. $\left(1 + \frac{n}{3}\right)$ b. $\left(1 + \frac{2}{n}\right)$ c. $\left(1 + \frac{n}{2}\right)$ d. $\left(1 + \frac{1}{n}\right)$

21. 4.0 g of a gas occupies 22.4 litres at NTP. The specific heat capacity of the gas at constant volume is $5.0 \text{ JK}^{-1} \text{ mol}^{-1}$. If the speed of sound in this gas at NTP is 952 ms^{-1} , then the heat capacity at constant pressure is (Take gas constant $R = 8.3 \text{ J/mol K}$): (2015 Re)

- a. 8.5 J/K mol b. 8.0 J/K mol
 c. 7.5 J/K mol d. 7.0 J/K mol

22. The amount of heat energy required to raise the temperature of 1 g of Helium at NTP, from $T_1 \text{ K}$ to $T_2 \text{ K}$ is: (2013)

- a. $\frac{3}{4} N_A k_B \left(\frac{T_2}{T_1} \right)$ b. $\frac{3}{8} N_A k_B (T_2 - T_1)$
 c. $\frac{3}{2} N_A k_B (T_2 - T_1)$ d. $\frac{3}{4} N_A k_B (T_2 - T_1)$

Mean Free Path

23. The mean free path for a gas, with molecular diameter and number density n can be expressed as : (2020)

- a. $\frac{1}{\sqrt{2} n \pi d^2}$ b. $\frac{1}{\sqrt{2} n^2 \pi d^2}$
 c. $\frac{1}{\sqrt{2} n^2 \pi^2 d^2}$ d. $\frac{1}{\sqrt{2} n \pi d}$

24. The mean free path ℓ for a gas molecule depends upon diameter, d of the molecule as (2020-Covid)

- a. $\ell \propto d$ b. $\ell \propto d^2$ c. $\ell \propto \frac{1}{d}$ d. $\ell \propto \frac{1}{d^2}$

25. The mean free path of molecules of a gas, (radius r) is inversely proportional to: (2014)

- a. r^3 b. r^2 c. r d. $\sqrt{r}$

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (a) | 3. (a) | 4. (b) | 5. (b) | 6. (d) | 7. (c) | 8. (d) | 9. (d) | 10. (c) |
| 11. (d) | 12. (b) | 13. (b) | 14. (b) | 15. (a) | 16. (c) | 17. (c) | 18. (c) | 19. (b) | 20. (b) |
| 21. (b) | 22. (b) | 23. (a) | 24. (d) | 25. (b) | | | | | |

